

W2 : CALCULUSOutline:-

Sets and functions

- Notation
- Logic
- Graphs and visualisation

Univariate Calculus

- Continuity and differentiability
- Derivatives and linear combos
- Application / Advanced Rules

Multivariate Calculus

- Lines and planes in high dimensional space
- Partial Derivate
- Gradients
- Linear approximations and Alternate gradient interpretation
- Application / Advanced Rules

Sets

$\mathbb{R} \Rightarrow$ Set of Real No.

$\mathbb{R}^+ \Rightarrow$ // // +ve Real no. + 0

$\mathbb{Z} \Rightarrow$ // // Integers

$\mathbb{Z}^+ \Rightarrow$ // // +ve // + 0

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$$

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}$$

$\mathbb{R}^d$: Set of d dimensional vectors = $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$ (d times)

$$\begin{pmatrix} 1 \\ 2 \\ 3.3 \end{pmatrix} \in \mathbb{R}^3$$

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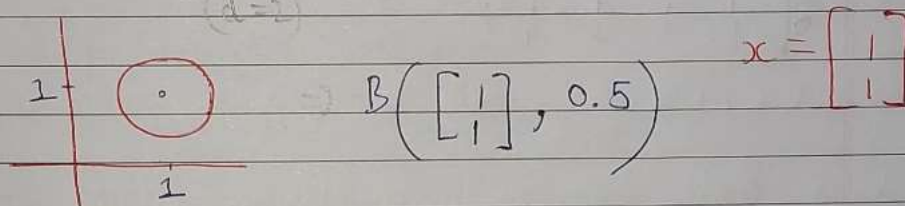
✓ $[a, b]^d = \{x \in \mathbb{R}^d : x_i \in [a, b] \text{ } i \in \{1, 2, \dots, d\}\}$

Metric Space:- Set + function distance

$\mathbb{R}^d : D(x, y) = \|x - y\|$
 $= \sqrt{(x_1 - y_1)^2 + \dots + (x_d - y_d)^2}$

$B(x, \epsilon) = \{y \in \mathbb{R}^d : D(x, y) < \epsilon\}$ (don't contain bound)

$\bar{B}(x, \epsilon) = \{y \in \mathbb{R}^d : D(x, y) \leq \epsilon\}$ (contain Bound)



Sets and Logic:-

$U = \text{Universal Set}$, $A^c = U \setminus A$

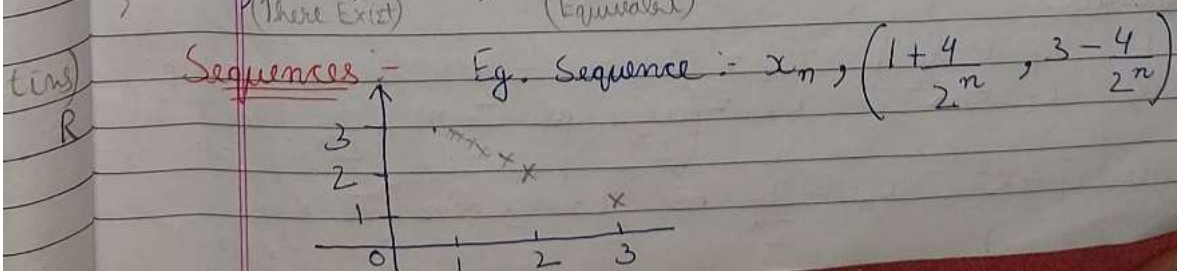
$(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$

$U = [0, 10]$, $A = [2, 5]$, $B = [4, 7]$

$A \cup B = [2, 7]$, $A \cap B = [4, 5]$, $(A \cup B)^c = [0, 2) \cup (7, 10]$

$A^c = [0, 2) \cup (5, 10]$, $B^c = [0, 4) \cup (7, 10]$

$\forall$, $\exists$, $\Rightarrow$, $\Leftrightarrow$
 (There Exist) , (Equivalent)



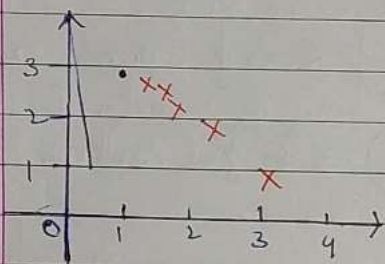
$x_1, x_2, \dots$

where $x_i \in \mathbb{R}^d$

$$\lim_{i \rightarrow \infty} x_i = x^* \Leftrightarrow \forall \epsilon > 0, \exists N \text{ s.t.}$$

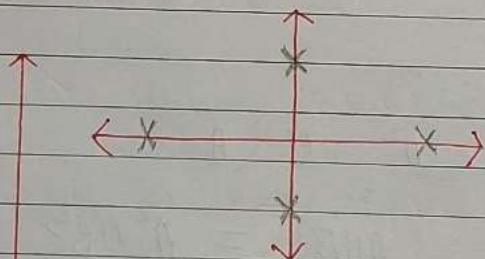
$$x_n \in B(x^*, \epsilon) \quad \forall n \geq N$$

Sequences:- Ordered collection of elements



Example Sequence:-

$$x_n = \left(1 + \frac{4}{2^n}, 3 - \frac{4}{2^n} \right)$$



$$x_n = \left(\cos \frac{\pi n}{2}, \sin \frac{\pi n}{2} \right)$$

(Non convergent)

We have, $x_1, x_2, \dots$ where $x_i \in \mathbb{R}^d$

$$\lim_{i \rightarrow \infty} x_i = x^* \Rightarrow \forall \epsilon > 0, \exists N \text{ s.t.}$$

(Ball with centre)

$$x_n \in B(x^*, \epsilon) \quad \forall n > N$$

Sequences:-

eg. i) $x_i \in \mathbb{R}$; $x_n = 1 + n$ (does this seq. converge)

ii) $x_i \in \mathbb{R}^2$; $x_n = \left(\frac{1}{2^n} \cos \left(\frac{\pi n}{2} \right), \frac{1}{2^n} \sin \left(\frac{\pi n}{2} \right) \right)$

iii) $x_i \in \mathbb{R}^2$; $x_n = \left(\frac{1}{2^n} \cos \left(\frac{\pi n}{2} \right), \sin \left(\frac{\pi n}{2} \right) \right)$

Vector Space :-

- If V is a Vector Space then :-
 $u \in V, v \in V$

$$\alpha, \beta \in \mathbb{R}$$

Any linear combo. of u and v is a Vector Space

$$\alpha u + \beta v \in V$$

$\mathbb{R}^d$ is a vector space

- $x \cdot y = x^T y = \sum_{i=1}^d x_i y_i$ (dot product)

- $\|x\|^2 = x \cdot x = x^T x = \sum_{i=1}^d x_i^2$

- x & y are perpendicular / orthogonal
 $x \cdot y = x^T y = \sum x_i y_i = 0$

Functions and Graphs

$$f: A \rightarrow B \quad \left(\begin{array}{l} \text{Real valued } f^n \text{ means co-domain} \\ \text{Domain} \quad \text{Codomain} \quad \text{- } n \text{ must be } \mathbb{R} \end{array} \right)$$

1-D funct :- $f: \mathbb{R} \rightarrow \mathbb{R}$

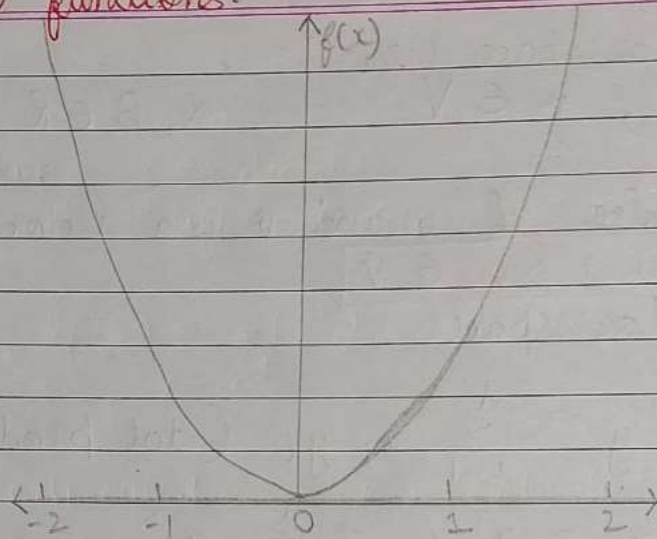
d-Dimensional or Multivariate functions

$$f: \mathbb{R}^d \rightarrow \mathbb{R}$$

Graph :- (C_f) $C_f \subseteq \mathbb{R}^{d+1}$

$$C_f = \{(x, f(x)) : x \in \mathbb{R}^d\}$$

Plots of 1D functions:-



$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2$$

$$G_f = \{(x, x^2) : x \in \mathbb{R}\} \in \mathbb{R}^2$$

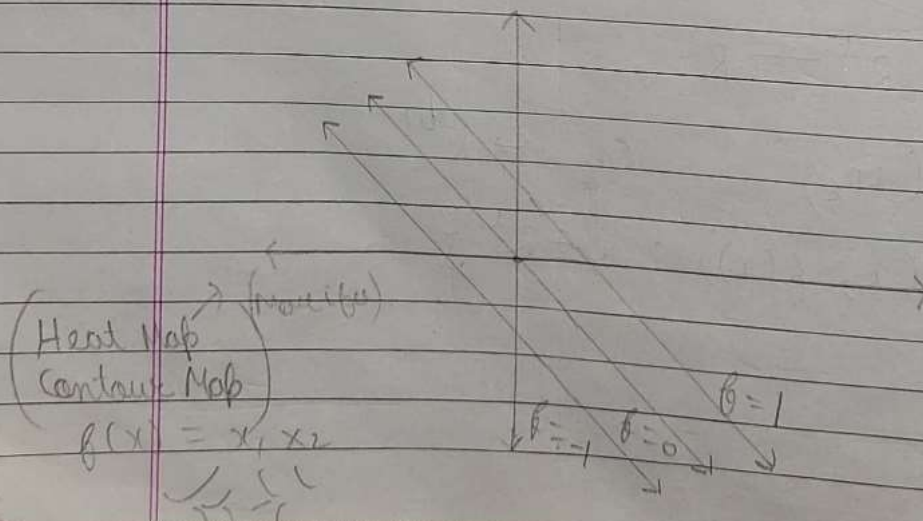
Contour Plots for 2D Functions:-

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}, \quad f(x) = x_1 + x_2$$

$$\text{Values} = \{-1, 0, 1\} \quad f(x) = -1$$

$$\Rightarrow x_1 = -x_2 - 1$$

$$f(x) = 0 \Rightarrow x_1 = -x_2$$



W2L2 Univariate Calculus : Continuity & Differentiability

Continuity of functions:- $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous at

$x^* \in \mathbb{R}$ if $\forall$ sequences $x_1, x_2, \dots$ converging to x^* .
We have that $f(x_i)$ converges to $f(x^*)$

$$\lim_{i \rightarrow \infty} x_i = x^* \Rightarrow \lim_{i \rightarrow \infty} f(x_i) = f(x^*)$$

$$\lim_{x \rightarrow x^*} f(x) = f(x^*)$$

eg 1:- $f(x) = x^2$, $x^* = 2$

eg seq is $x_i: 3, 2.5, 2.25, \dots \rightarrow 2$
 $f(x_i): 9, 6.25, 2.25^2 \dots \rightarrow 4$

eg 2) $f(x) = \text{sign}(x) = \begin{cases} -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 1 & \text{if } x > 0 \end{cases}$
 $x^* = 0$

$x_i: 1, \frac{1}{2}, \frac{1}{4} \rightarrow 0$

$f(x_i): 1, 1, 1 \rightarrow 1$

$x_i: -1, -\frac{1}{2}, -\frac{1}{4} \rightarrow 0$

$f(x_i): -1, -1, -1 \rightarrow -1$

(Funct. is cont. if it is continuous at all parts in its domain)

eg 3): $f(x) = \begin{cases} 2x+1 & \text{if } x > 1 \\ 3 & \text{if } x \leq 1 \end{cases}$

eg 4) $f(x) = \frac{1}{x}$

eg: 3 $f(x) = \sin\left(\frac{1}{x}\right)$

Differentiability of Functions:

A fⁿ $f: \mathbb{R} \rightarrow \mathbb{R}$ is differentiable at $x^* \in \mathbb{R}$ if

$$f'(x) = \lim_{x \rightarrow x^*} \frac{f(x) - f(x^*)}{x - x^*} \text{ exists}$$

f is not continuous at $x^* \Rightarrow f$ is not differentiable

eg:- $f(x) = |x|$

$$x_i: 1, \frac{1}{2}, \frac{1}{4}, \dots \rightarrow 0$$

$$\frac{f(x_i) - f(0)}{x_i} = 1, 1, 1, \dots \rightarrow 1$$

eg 2 $\rightarrow x_i: -1, -\frac{1}{2}, -\frac{1}{4}, \dots \rightarrow 0$

$$\frac{f(x_i) - f(0)}{f(x_i)} = -1, -1, -1, \dots$$

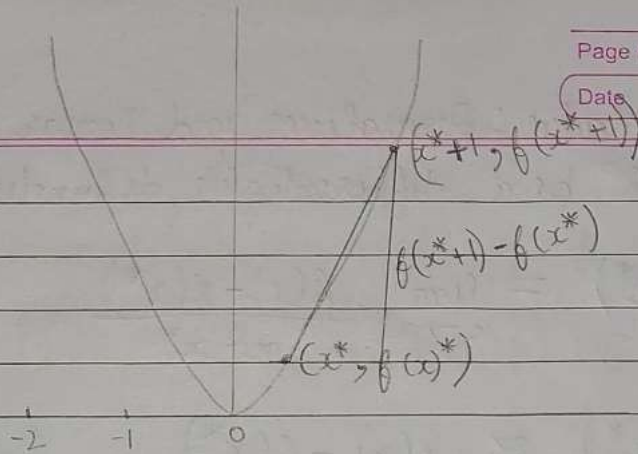
eg. 2 $\rightarrow f(x) = \begin{cases} 4x+2 & \text{if } x \geq 2 \\ 2x+8 & \text{if } x < 2 \end{cases}$

Since it isn't continuous at $x^* = 2$
 $\Rightarrow$ It is also not differentiable

eg 3 $\rightarrow f(x) = \begin{cases} 4x+2 & \text{if } x \geq 2 \\ 2x+6 & \text{if } x < 2 \end{cases}$

$$\lim_{x \rightarrow 2^+} \frac{f(x) - f(2)}{x - 2} = 4$$

$$\lim_{x \rightarrow 2^-} \frac{f(x) - f(2)}{x - 2} = 2$$



~~Suppose~~ $f'(x^*) = \lim_{x \rightarrow x^*} \frac{f(x) - f(x^*)}{x - x^*}$ Or

$$f'(x^*) = \lim_{h \rightarrow 0} \frac{f(x^* + h) - f(x^*)}{h}$$

AQ 2.2 1) $|x| = \begin{cases} -x, & x \leq 0 \\ x, & x \geq 0 \\ 0, & x = 0 \end{cases}$

Univariate Calculus: Derivatives and Linear Approximations

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function

$$f'(x^*) = \lim_{x \rightarrow x^*} \frac{f(x) - f(x^*)}{x - x^*}$$

$$f(x) \approx \frac{f(x) - f(x^*)}{x - x^*} \quad (\text{around } x = x^*)$$

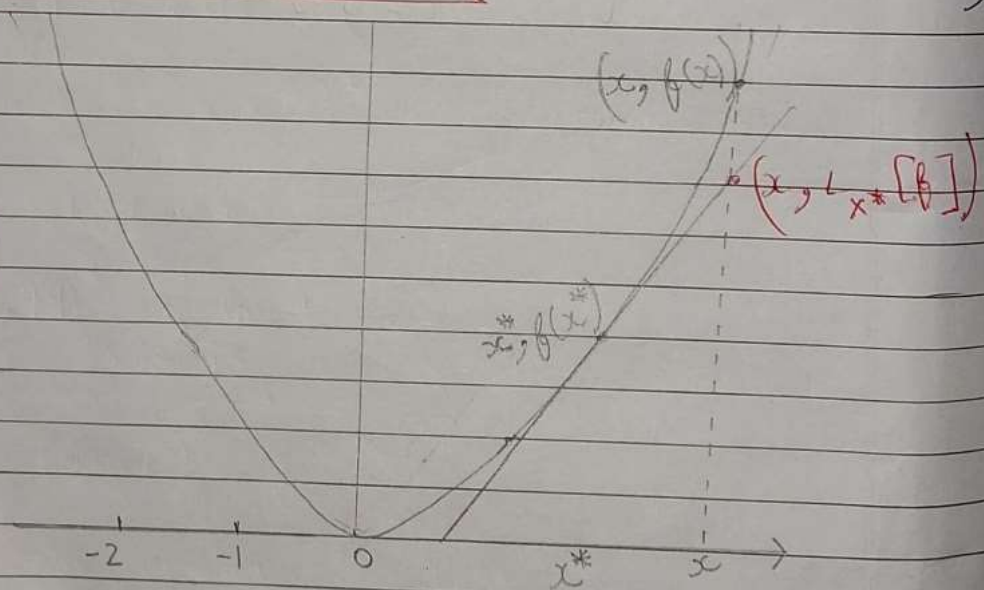
(Linear Approx.)

$$f(x) \approx f(x^*) + f'(x^*) (x - x^*) \quad (\text{around } x = x^*)$$

$$L_{x^*}[f](x)$$

$$f(x) \approx L_{x^*}[f](x) \quad (\text{around } x = x^*)$$

$$(f(x) = x^2)$$



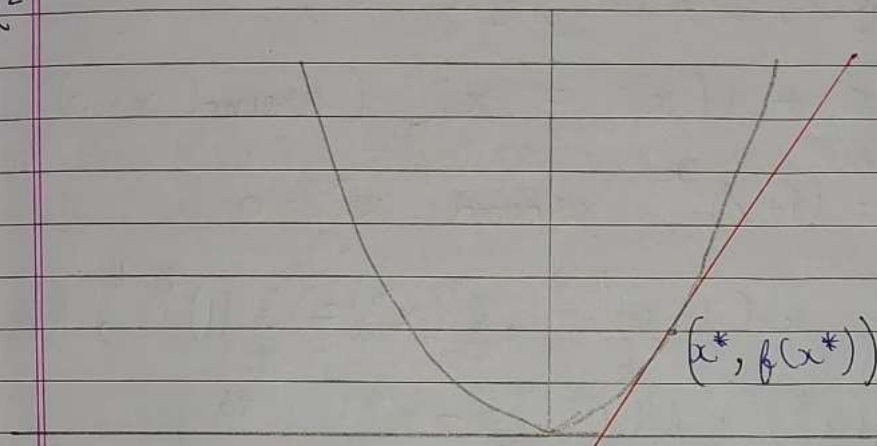
$$L_{x^*}[f] = f(x^*) + f'(x^*) (x - x^*)$$

$$= 1^2 + 2(x - 1) = 1 + 2x - 2$$

$$= 2x - 1 \quad (\text{around } x = 1)$$

Linear Approximation and Tangent lines

$L_{x^*}[f]$ is a tangent to the graph of f at the point $x^*, f(x^*)$



eg. i) Linear Approx of $f(x) = \sin(x)$ around $x^* = 0$

$$f'(x) = \cos(x) \quad f'(x^*) = 1 \quad f(x^*) = 0$$

$$f(x) \approx f(x^*) + f'(x^*)(x - x^*)$$

$$= 0 + 1(x - 0) = x \quad (\text{around } x = 0)$$

$$\Rightarrow \sin x \approx x \quad \text{if } x \approx 0$$

ii) $f(x) = e^x$ around $x^* = 0$

$$f'(x) = e^x, \quad f'(x^*) = 1, \quad f(x^*) = 1$$

$$f(x) \approx f(x^*) + f'(x^*)(x - x^*)$$

$$= 1 + 1(x) = x + 1 \quad (\text{around } x = 0)$$

iii) $\ln(1+x)$ around $x^* = 0$

$$f'(x) = \frac{1}{1+x}, \quad f'(x^*) = 1, \quad f(x^*) = \ln(1)$$

$$f(x) \approx f(x^*) + f'(x^*)(x - x^*)$$

$$\approx 0 + 1(x) = x \quad (\text{around } x=0)$$

iv) $f(x) = (1+x)^x$ around $x^* = 0$

$$f'(x) = x(1+x)^{x-1}, \quad f'(x^*) = x(1)^{x-1} = x, \quad f(x^*) = 1$$

$$f(x) \approx f(x^*) + f'(x^*)(x - x^*)$$

$$\approx 1 + x(1)^{x-1}(x - 0)$$

$$\approx 1 + x x(1)^{x-1} \quad (\text{around } x=0)$$

$$\approx 1 + x x$$

→ DMP

v) (0.99) is closest to

a) 0.95 b) 0.93 c) 0.91 , d) 0.9

$$f'(x) = 7(0.99)^6, \quad f'(x^*) =$$

(Use above formula $1 + x x$)

$$\Rightarrow 1 + -0.01 \times 7 = 0.93$$

AQ 2.3 Higher order approx. are more accurate than linear approx. ? True

2) Plot of L.A. of a funct. at a pt. it is tangent to f at the pt. $(v, f(v))$

3) length of vector $[1 \ 2 \ 3]$ is?

$$\sqrt{(1)^2 + (2)^2 + (3)^2} = \sqrt{14} =$$

4) What is perp. to any given vector?

Univariate Calculus: Applications and Advanced Rules Higher Order Approximations:-

$$f(x) \approx L_{x^*}[f](x) \quad (\text{around } x = x^*)$$

$$f(x) \approx f(x^*) + f'(x^*)(x - x^*) \quad (\text{Linear}) \text{ Approx}$$

$$f(x) \approx f(x^*) + f'(x^*)(x - x^*) + \frac{1}{2} f''(x^*)(x - x^*)^2$$

✓ Quadratic Approx is more accurate but complex to calculate.

eg 1 $\rightarrow f(x) = x^2$
Ans $\rightarrow f'(x) = 2x$
 $f''(x) = 2$

$$x^2 \approx (x^*)^2 + 2x^*(x - x^*) + \frac{1}{2} \cdot 2 \cdot (x - x^*)^2$$

$$= x^2$$

eg 2) e^x around $x^* = 0$

$$f'(x) = e^x \quad f'(x^*) = 1 \quad f(x^*) = 1$$

$$e^x \approx 1 + 1(x) + \frac{1}{2} x e^0 x (x - 0)$$

$$\approx x + 1 + \frac{x^2}{2}$$

Q-2) Closest to $(1.1)^7 \Rightarrow$

$$f = (1+x)^7, \quad f'(x) = 7(1+x)^6$$

$$f''(x) = 42(1+x)^5, \quad f(0) = 1$$

$$f'(0) = 7, \quad f''(0) = 42$$

$$f(1.1) = 1 + 7(0.1) + \frac{1}{2} \cdot 42(0.01)$$

$$= 1.91$$

Product Rule :-

$$f(x) = g(x) h(x) \quad f'(x) = ?$$

$$f(x) \approx (g(0) + x g'(0))(h(0) + x h'(0))$$

$$= g(0) h(0) + x [g'(0) h(0) + h'(0) g(0)]$$

$$+ x^2 g'(0) h'(0)$$

→ (I can ignore quad)

$$L_x[f] = f(0) + x f'(0)$$

$$f'(0) = g'(0) h(0) + h'(0) g(0) \quad \checkmark$$

Chain Rule :-

$$f(x) = g(h(x))$$

$$= g(h(0) + h'(0)x) = g(h(0)) + g'(h(0)) [h(0) + h'(0)x - h(0)]$$

$$= g(h(0)) + g'(h(0)) h'(0)x$$

$$f'(0) = g'(h(0)) h'(0) \quad \checkmark$$

eg. 1) $\frac{e^{3x}}{\sqrt{1+x}}$ Lim. approx at $x=0$

$(1+x)^n = 1 + nx$ (Im LA) using same for den & num

$$e^{3x} = (1+3x) \quad \left| \quad (1+x)^{-1/2} = 1 - \frac{x}{2} \right.$$

$$= (1+3x) \left(1 - \frac{x}{2}\right) \quad \left(\text{Remove quad. part} \right)$$

$$= 1 + 3x - \frac{x}{2} - \frac{3x^2}{2} = 1 - \frac{5x}{2}$$

ii) $e^{\sqrt{1+x}}$ around $x=1$

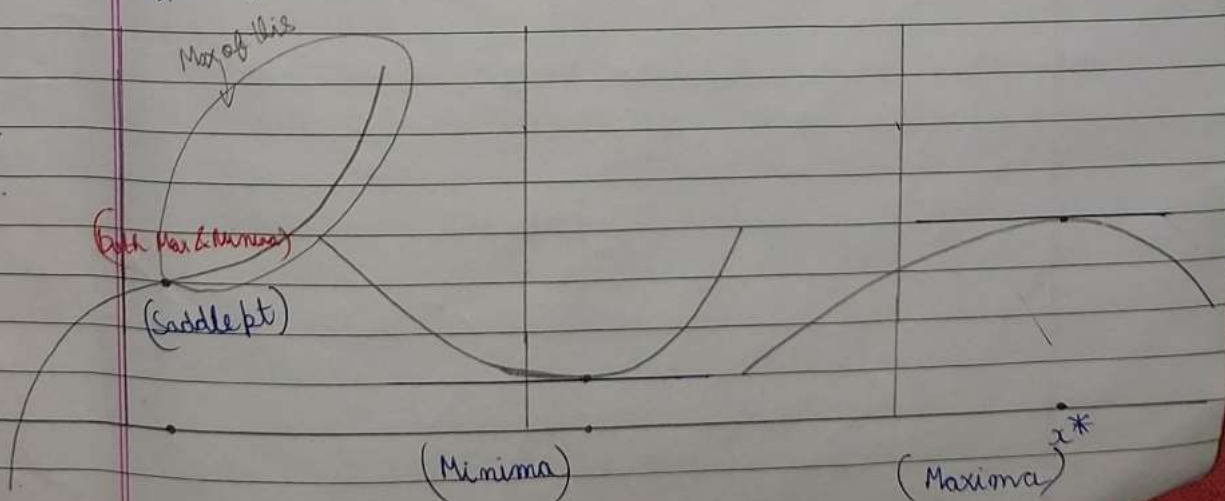
$$\approx e^{\sqrt{2}} + \frac{e^{\sqrt{2}}}{2\sqrt{2}} (x-1)$$

Maxima, Minima & Saddle pt.:-

$$L_{x^*}[f] = f(x^*) + f'(x^*)(x-x^*)$$

$$f'(x^*) = 0 \quad \Leftrightarrow \quad x^* \text{ is a critical pt. of } f.$$

$$L_{x^*}[f] = 7 + 3x$$



AQ 2.4

1) Representation of a multivariate function?

Ans 1 $f: \mathbb{R}^d \rightarrow \mathbb{R}$

2) Hyperplane normal to a vector w with offset value b given by $\{x: w^T x = b\}$

3) Critical pt. for $f(x) = x^2 - x$ is?

$$f'(x) = 2x - 1, \quad \cancel{f'(0) = -1}, \quad \cancel{f(0) = 0}$$

$$\cancel{f(x) \approx 0 - 1(x - 0)} \quad x = 0.5$$

4) $f(x) = x^2 - x$

$\Rightarrow$ Minima

Multivariate Calculus: Lines and Planes in higher dimensional space

Geometry of Lines:-

i) A line in $\mathbb{R}^d \subseteq \mathbb{R}^d$

ii) A line through the point $v \in \mathbb{R}^d$ along the vector $v \in \mathbb{R}^d$

$$= \{x \in \mathbb{R}^d : x = v + \alpha v \text{ for } \alpha \in \mathbb{R}\}$$

b) Line through $v, v' \in \mathbb{R}^d$

$$= \{x \in \mathbb{R}^d : x = v + \alpha(v' - v) \text{ for } \alpha \in \mathbb{R}\}$$

$$= \{x \in \mathbb{R}^d : x = (1-\alpha)v + \alpha v' \text{ for } \alpha \in \mathbb{R}\}$$

AKA Line through v along $v' - v$
 // // v' // $v - v'$

Geometry of Hyperplane

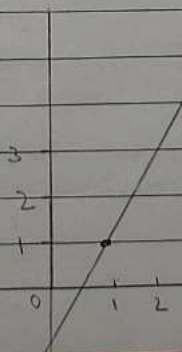
A $(d-1)$ dimensional hyperplane $\subseteq \mathbb{R}^d$

A hyperplane normal to vector $w \in \mathbb{R}^d$ with value $b \in \mathbb{R}$

$$= \{x \in \mathbb{R}^d : w^T x = b\}$$

$$= \left\{x \in \mathbb{R}^d : \sum_{i=1}^d w_i x_i = b\right\}$$

Example Lines:-



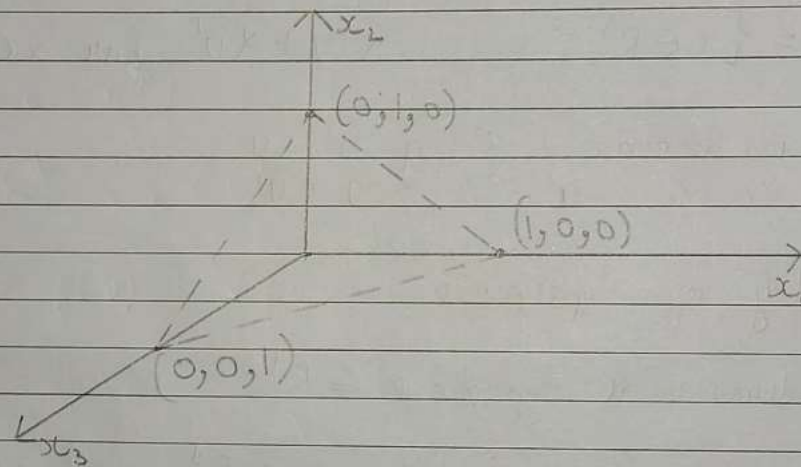
Line through $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ along $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$$\{x \in \mathbb{R}^2 : x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 2 \end{bmatrix} ; \lambda \in \mathbb{R}\}$$

eg. plane:- $d=3$

Hyperplane normal / perp / orthogonal to $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ with value 1.

$$= \{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 1\}$$



The pt. $(0, 1, 0)$ lies on T, which is perp. to vector $(1, 1, 1)$

Tuples vs Points vs Vectors
 (Programming (Location) (Direction)
 lang)

Partial Derivatives:-

$$f: \mathbb{R}^2 \rightarrow \mathbb{R} \quad f(x_1, x_2) = x_1^2 + x_2^2$$

$$\frac{\partial f}{\partial x_1}(v) = \frac{f(v) + \begin{bmatrix} \alpha \\ 0 \end{bmatrix} - f(v)}{\alpha}$$

$$= \frac{f(v_1 + \alpha, v_2) - f(v_1, v_2)}{\alpha}$$

$$\frac{\partial f}{\partial x_2}(v) = \frac{f(v_1, v_2 + \alpha) - f(v_1, v_2)}{\alpha}$$

$$\frac{\partial f}{\partial x_i}(v) = \lim_{\alpha \rightarrow 0} \frac{f(v + \alpha e_i) - f(v)}{\alpha}$$

Gradient

$$f: \mathbb{R}^d \rightarrow \mathbb{R}$$

$$\frac{\partial f}{\partial x} = \left[\frac{\partial f}{\partial x_1}(v), \frac{\partial f}{\partial x_2}(v), \dots, \frac{\partial f}{\partial x_d}(v) \right]$$

$$\nabla f(v) = \left[\frac{\partial f}{\partial x} \right]^T$$

eg) $d=2$ $f(x) = x_1^2 + x_2^2$

$$\frac{\partial f}{\partial x_1}(v) = 2x_1, \quad \frac{\partial f}{\partial x_2}(v) = 2x_2 \quad \Rightarrow \nabla f(v) = \begin{bmatrix} 2v_1 \\ 2v_2 \end{bmatrix}$$

$\Rightarrow f: \mathbb{R}^3 \rightarrow \mathbb{R} \quad f(x) = x_1 + 2x_2 + 3x_3$

$$\nabla f(v) = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

(Linear Function)

AQ 2.5

1) Line through point u along vector v is :-

$$x: x = u + \alpha v$$

2) Gradient of a funct. evaluated at a pt is vector.

True

2.6 1 Gradients and Linear Approximation:-

$$f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) \approx f(x^*) + f'(x^*)(x - x^*)$$

(around $x = x^*$)

$$L_{x^*}[f](x)$$

$$v \in \mathbb{R}^d, x \in \mathbb{R}^d$$

$$f(x) \approx f(v) + \nabla f(v)^T (x - v)$$

$$= f(v) + \sum_{i=1}^d \frac{\partial f}{\partial x_i}(v) \cdot (x_i - v_i)$$

$$L_v[f](x)$$

(around $x = v$)Lin. Approx. of f around pt. v evaluated at x and it is valid at $x = v$ eg. 1)

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$f(y_1, v_2) \approx f(v_1, v_2)$$

$$+ \frac{\partial f}{\partial x_1}(v)(y_1 - v_1)$$

$$f(y_1, v_2) - f(v_1, v_2) \approx \frac{\partial f}{\partial x_1}(v)(y_1 - v_1)$$

$$f(v_1, y_2) - f(v_1, v_2) \approx \frac{\partial f}{\partial x_2}(v)(y_2 - v_2)$$

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Date / /

What if both y_1, y_2 move simultaneously

$$f(y_1, y_2) - f(v_1, v_2) = \frac{\partial f}{\partial x_1}(v)(y_1 - v_1) + \frac{\partial f}{\partial x_2}(v)(y_2 - v_2)$$

$$f(y_1, y_2) \approx f(v_1, v_2) + \nabla f(v)^T (y - v)$$

eg. $\rightarrow f(x_1, x_2) = x_1^2 + x_2^2$

$\Rightarrow$ i) Approximate f around $(6, 2)$

$$f(v) = 40, \quad \nabla f(v) = \begin{bmatrix} 12 \\ 4 \end{bmatrix} \quad \left(\text{Since } \nabla f(x) = \begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix} \right)$$

$$\Rightarrow f(x) \approx 40 + \begin{bmatrix} 12 & 4 \end{bmatrix} \begin{bmatrix} x_1 - 6 \\ x_2 - 2 \end{bmatrix}$$

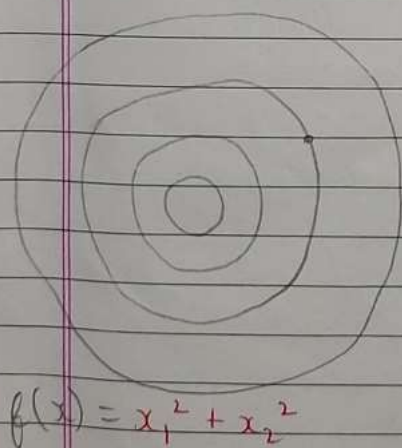
$$= 40 + 12(x_1 - 6) + 4(x_2 - 2)$$

$$= 40 + 12x_1 + 4x_2 - 72 - 8$$

$$= 12x_1 + 4x_2 - 40$$

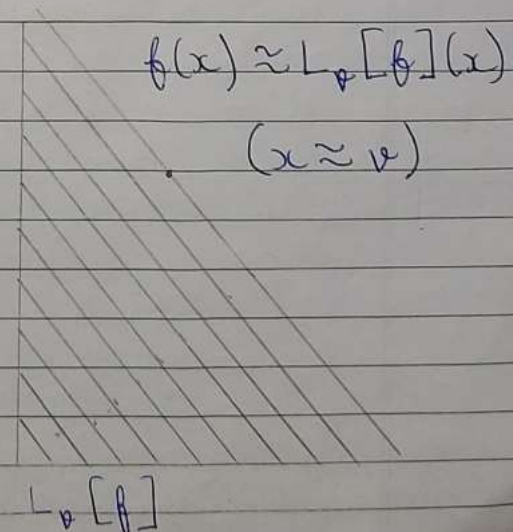
around $(x_1, x_2) \approx (6, 2)$

Contour Plot for above function



$$f(x) = x_1^2 + x_2^2$$

L.A $\rightarrow$



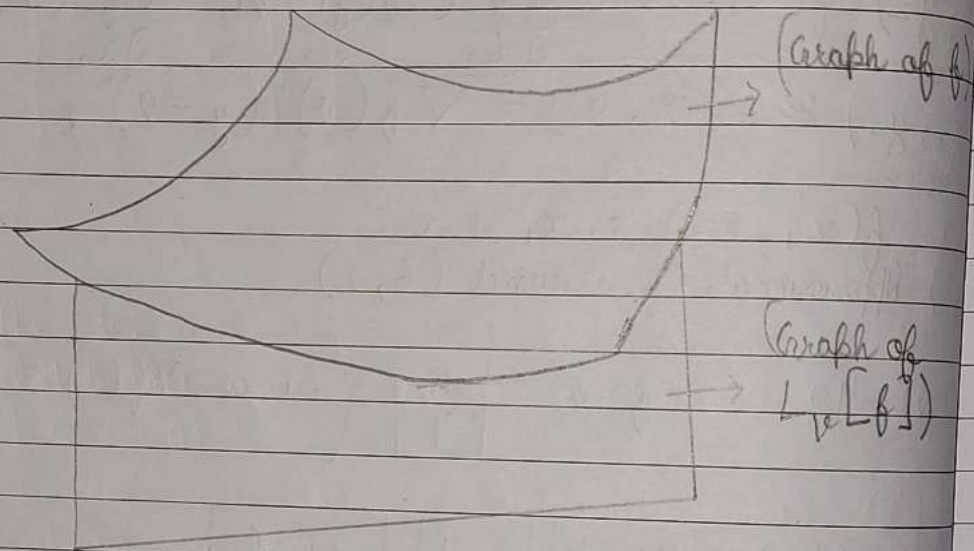
$$f(x) \approx L_v[f](x)$$

$$(x \approx v)$$

$$L_v[f]$$

Gradient and Tangent Plane

The graph of $L_v[f]$ is a plane that is tangent to the graph of f at point $(v, f(v))$



Gradient & Contours

$$\nabla f(v) \perp \{x \in \mathbb{R}^d : f(x) = f(v)\}$$

$$v = \begin{bmatrix} -6 \\ 2 \end{bmatrix}$$

$$\nabla f(v) = \begin{bmatrix} -12 \\ 4 \end{bmatrix}$$

Notes By Hardik Sharma

If any part is blurred or not understandable
message me on WhatsApp (9877100914)

Tip: Don't directly jump onto the notes firstly
watch the whole video and then read them.